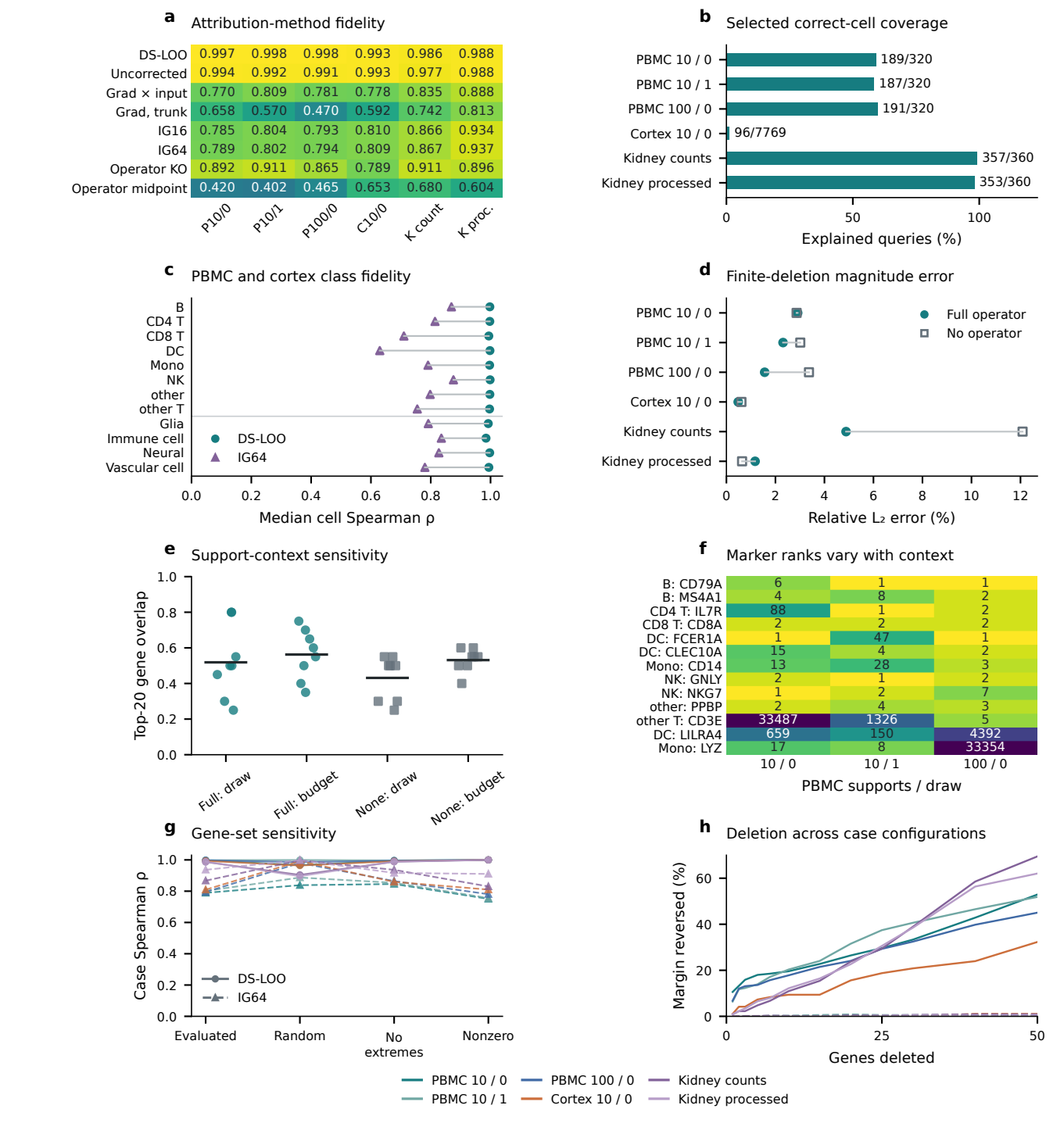

a
Attribution-method fidelity
DS-LOO 0.997 0.998 0.998 0.993 0.986 0.988
Uncorrected 0.994 0.992 0.991 0.993 0.977 0.988
Grad × input 0.770 0.809 0.781 0.778 0.835 0.888
Grad, trunk 0.658 0.570 0.470 0.592 0.742 0.813
IG16 0.785 0.804 0.793 0.810 0.866 0.934
IG64 0.789 0.802 0.794 0.809 0.867 0.937
Operator KO 0.892 0.911 0.865 0.789 0.911 0.896
Operator midpoint 0.420 0.402 0.465 0.653 0.680 0.604
P10/0
P10/1
P100/0
C10/0
K count
K proc.
b
Selected correct-cell coverage
PBMC 10 / 0
PBMC 10 / 1
PBMC 100 / 0
Cortex 10 / 0
Kidney counts
Kidney processed
189/320
187/320
191/320
96/7769
357/360
353/360
0
50
100
Explained queries (%)
c
PBMC and cortex class fidelity
B
CD4 T
CD8 T
DC
Mono
NK
other
other T
Glia
Immune cell
Neural
Vascular cell
DS-LOO
IG64
0.0
0.2
0.4
0.6
0.8
1.0
Median cell Spearman ρ
d
Finite-deletion magnitude error
PBMC 10 / 0
PBMC 10 / 1
PBMC 100 / 0
Cortex 10 / 0
Kidney counts
Kidney processed
Full operator
No operator
0
2
4
6
8
10
12
Relative L₂ error (%)
e
Support-context sensitivity
Top-20 gene overlap
0.0
0.2
0.4
0.6
0.8
1.0
Full: draw
Full: budget
None: draw
None: budget
f
Marker ranks vary with context
B: CD79A 6 1 1
B: MS4A1 4 8 2
CD4 T: IL7R 88 1 2
CD8 T: CD8A 2 2 2
DC: FCER1A 1 47 1
DC: CLEC10A 15 4 2
Mono: CD14 13 28 3
NK: GNLY 2 1 2
NK: NKG7 1 2 7
other: PPBP 2 4 3
other T: CD3E 33487 1326 5
DC: LILRA4 659 150 4392
Mono: LYZ 17 8 33354
10 / 0
10 / 1
100 / 0
PBMC supports / draw
g
Gene-set sensitivity
Case Spearman ρ
0.0
0.2
0.4
0.6
0.8
1.0
DS-LOO
IG64
Evaluated
Random
No extremes
Nonzero
h
Deletion across case configurations
Margin reversed (%)
0
20
40
60
0
25
50
Genes deleted
PBMC 10 / 0
PBMC 10 / 1
PBMC 100 / 0
Cortex 10 / 0
Kidney counts
Kidney processed